MIL-R-27208/8F
22 October 1970
SUPERSEDING
MIL-R-27208/8E
2 July 1970
(See 6.3)

MILITARY SPECIFICATION

RESISTORS, VARIABLE, WIRE-WOUND

(ADJUSTMENT TYPE, LEAD-SCREW ACTUATED),

STYLE RT12

THIS SPECIFICATION INACTIVE FOR DESIGN AFTER DATE OF THIS SPECIFICATION (SEE 6.3). USE MIL-R-39015/1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirements for style RT12, adjustment type, lead-screw actuated, wire-wound, variable resistors.

2. APPLICABLE DOCUMENTS

2.1 The following document of the issue in effect on date of invitation for bids or request for proposal forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-R-27208 - Resistors, Variable, Wire-Wound (Adjustment Type), General Specification for.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-27208, and as specified herein.

3.2 Design and construction. Resistors shall be of the design, construction, and physical dimensions specified on figure.

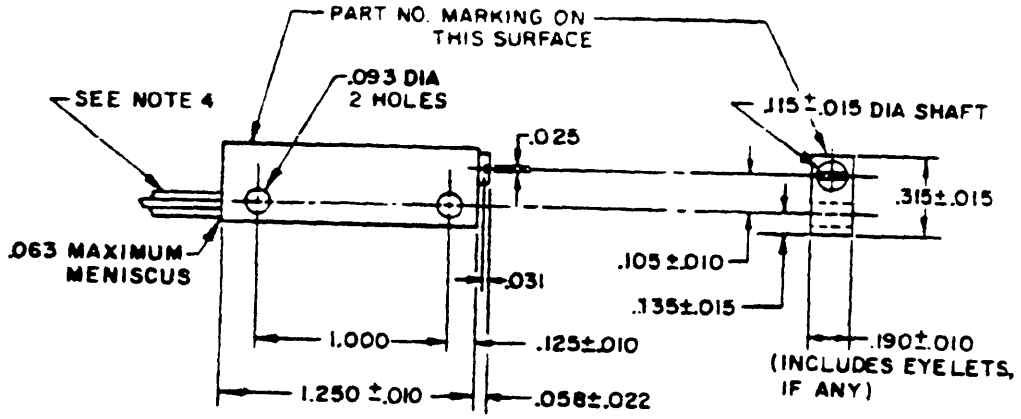
3.3 Power rating. The power rating shall be 3/4 watt, based on full-load operation at an ambient temperature of 85° C.

3.4 Nominal resistance value, maximum resolution, and maximum rated ac or dc working voltage. Nominal resistance values, maximum resolutions, and maximum rated ac or dc working voltages are as specified in table I.

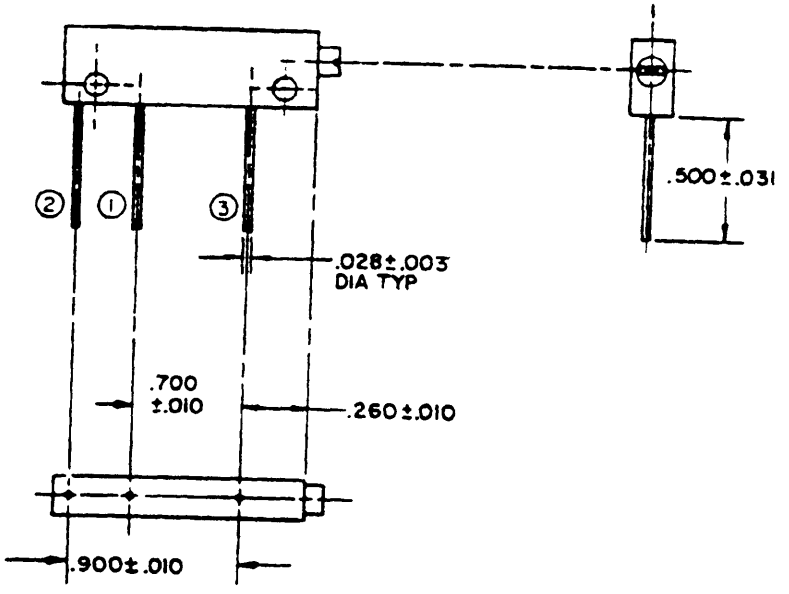
3.5 Actual effective electrical travel. Actual effective electrical travel shall be 17 turns minimum, and 27 turns maximum.

3.6 Operating torque. Operating torque shall be a maximum of 8 ounce-inches.

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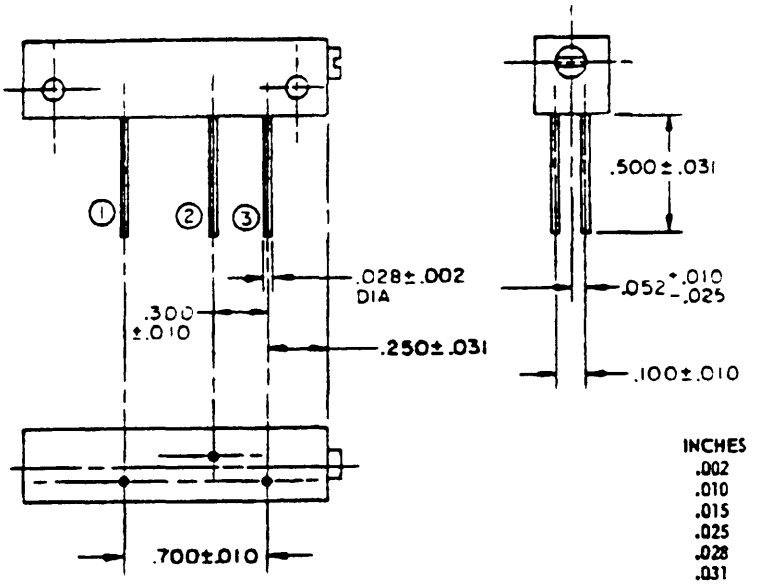
FLEXIBLE LEAD TERMINAL TYPE - L



PRINTED-CIRCUIT PIN TERMINAL TYPE-P

FIGURE 1. Style RT12 resistors.

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PRINTED-CIRCUIT
PIN TERMINAL TYPE - Y

INCHES	MM
.002	.05
.010	.25
.015	.38
.025	.64
.028	.71
.031	.79
.052	1.32
.058	1.47
.063	1.60
.093	2.36
.100	2.54
.105	2.67
.115	2.92
.125	3.18
.135	3.43
.190	4.83
.250	6.35
.260	6.60
.300	7.62
.315	8.00
.500	12.70
.700	17.78
.900	22.86
1.000	25.40
1.250	31.75

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.005 (.13 mm).
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. The three leads shall be of stranded wire, AWG size 28 to 30, having a minimum length of 6 (152.4 mm); they shall be insulated with polytetrafluoroethylene, stripped .250 ±.062 (6.35 ±1.57 mm) from the end, and color coded.

FIGURE 1. Style RT12 resistors - Continued.

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TABLE I. Nominal resistance value, maximum resolution, and maximum rated ac or dc working voltage.

Nominal resistance value (in ohms)	Maximum resolution 3/	Maximum rated ac or dc working voltage
Ohms	Percent	Volts
10	2.2	2.7
20	2.0	3.8
50	1.3	6.1
100	1.1	8.7
200	0.9	12.3
500	0.6	19.4
1,000	0.5	27.4
2,000	0.4	38.7
5,000	0.3	61.3
10,000 1/	0.3	86.7
20,000 2/	0.3	122.0

1/ Value based on the use of wire having no less than a 0.001-inch nominal diameter.
2/ Value based on the use of wire having no less than a 0.0008-inch nominal diameter.
3/ Maximum resolutions shown are theoretical.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection shall be in accordance with MIL-R-27208.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance wtih MIL-R-27208.

6. NOTES

6.1 The notes specified in MIL-R-27208 are applicable to this specification.

* 6.2 Supersession data. This specification supersedes MIL-R-27208/8E, 2 July 1970, and MIL-R-27208/3E, 19 December 1969. Style RT12, terminal types L and Y of this specification, are substitutes for style RT11 resistors having the same resistance values as terminal types L and P, respectively, of MIL-R-27208/3E.

6.3 Substitution data. The resistors specified herein are not for use in design after date of this specification. They are authorized for use in design contracts effective prior to or on the date of this specification, and to support existing military equipment. Resistors specified in MIL-R-39015/1 are preferred for design and regardless of the failure rate designation can be used as substitutes for the inactivated resistors of the same resistance value, tolerance and characteristic performance of this specification.

6.4 Weight. The maximum weight is 4.5 grams.

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - MI, MU, EL
Navy - EC, AS, OS
Air Force - 11, 17, 80
DSA - ES

User activities:
Army - ME, AT, AV
Navy - MC
Air Force - 19

Preparing activity:
Air Force - 80

Agent:
DSA - ES

(Project 5905-0733)

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